



HTC announces Link VR

Link VR is the first mobile headset to offer six-degrees-of-freedom (6DOF) tracking for its U11 phone. <http://dgit.in/HTClink>



Amazon's shoppable stickers

Amazon's mobile app adds stickers allowing users to view and purchase objects using the phone's camera. <http://dgit.in/AmazonCam>

RNA is the prime candidate for the molecule that existed in the primordial soup that first began creating copies of itself while also having the ability to synthesise proteins.

Unfortunately, after decades of research, no one was able to synthesise nucleotides (the molecules that make RNA), no trace of self-replicating RNA has been found in nature, and no one has been able to explain how it might have first emerged in the primordial soup. Despite its shortcomings, the RNA World theory remains one of the most important theories that attempts to explain abiogenesis. It may not have



I have a joke about DNA but you'll just copy it.

been able to solve the mystery, but it certainly laid the groundwork for further research, even other theories depend heavily on the findings from RNA World while constructing newer models for the origin of life.

While the RNA World theory had hit a dead end, another was picking up steam, centred around the idea that before life self-replicates, it must first know how to harness energy. We now know that when a cell needs energy, it breaks a third phosphate off the molecule adenosine triphosphate (ATP) – releasing energy in the process. This is how cells store and release energy. But how exactly do cells make ATP?

Biochemist Peter Mitchell, who later won the 1978 Nobel Prize in Chemistry, supplied the answer: The cells were pumping charged particles called proton across the membrane to supply the energy, in other words, cells have a proton gradient.

Geologist Mike Russell, Deborah Kelley of the University of Washington, and biologist William Martin found a spot with natural proton gradient: Deep

sea alkaline vents with the chemical energy needed to first make simple hydrocarbons and then complex molecules like RNA. This model provided a viable alternative to primordial soup for explaining abiogenesis. Unfortunately, this is where we hit another roadblock in our journey. Critics maintain that the reactions needed for the deep sea vent theory to work are actually incompatible with water. The RNA world and deep sea vent theories presented compelling ideas about abiogenesis, unfortunately, neither theories could reach a fitting conclusion. This is when Pier Luigi Luisi of Roma Tre University presented an alternative based on the idea of self-preservation.

His reasoning was simple but hard to refute: How can you have either metabolism or self-replication unless you first have a container to keep the cell structures secure? In the biological world, this task is accomplished by the cell membrane – a barrier between the organelles and the outside environment – made from two layers of insoluble fatty acids called lipids. Jack Szostak, who previously worked on the RNA World hypothesis, teamed up with Luisi to create the crude predecessor to the biological cell, essentially blobs of lipid that they called the protocell. Then, they injected RNAs into protocells and studied their behavior.

The Lipid World theory had some spectacular successes. First, it demonstrated that protocells can split without spilling out their molecular guts, mirroring the process of fission in biological cells. Second, by adding a type of clay called montmorillonite, the vesicles formed hundred times faster, meaning, the protocells now had genes in the form of RNA and a catalyst in the form of clay, all in one neat lipid enclosure. This is the closest we have come to recreating a biological cell in the laboratory from scratch.

All the theories so far tried to build the model of life on top of essential biological processes, but each missed the mark in explaining the larger picture. John Sutherland of the Laboratory of

Molecular Biology in Cambridge UK had been following abiogenesis closely. He thought if he could just synthesise RNA nucleotides in the lab, we would get closer to finding the answer.

Each RNA nucleotide is made of a sugar, a base, and a phosphate, but sugar and base don't pair up easily owing to their different molecular shape. Sutherland's breakthrough was using cyanamide as a catalyst to induce an alternate chain of reactions, by doing this he was able to produce two of the four RNA nucleotides. It was a home run.

Sutherland was still skeptical of the RNA World, it seemed like an oversimplification to him. RNA was an important part, but not the holy grail. So he tried to build a more cohesive theory that took the best aspects of the previous ones and presented something more complex than the primordial soup: A place where cells could self-assemble all at once. According to Sutherland, there might have been a lake somewhere exposed to UV radiation that contained a hodgepodge of chemicals including water, hydrogen, ammonia, methane, phosphates, clay, minerals and catalysts, precursors to lipids... the messier this concoction, the better, as it allows for more possibilities in chemical synthesis. Then perhaps one day, a rudimentary first cell assembled itself from this chemical soup, and thrived in presence of suitable environment and absence of natural predators. This may seem too far out but it's more plausible in its complexity in relation to previous theories.

As you may have guessed by now, it's exceedingly hard to either fully accept one theory of abiogenesis or fully write off another one. RNA World, Deep Sea Vents, Lipid World, and Hodgepodge World all attempt to reveal the formative stages of the intricate machinery that goes into the making of every single living cell today, and thus, life itself.

Space may be the final frontier, but even here at home, there are puzzles still begging to be pieced together. In the meantime, we can marvel at the staggering complexity and sheer improbability of the chain of events that led us here. 